



Conjugate Logical Bases

A lateral attack on the quantum measurement problem

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vs 1.9



Argument

- Observation:
 - Logical self-reference requires *imaginary* truthvalues.
 - The Imaginaries are *conjugate* to the Booleans.
 - A form *paradoxical* in one basis will be *indeterminate* in the other.
- Speculation:
 - A self-referential quantum system can *self-collapse* in the one basis in which it is indeterminate.
 - No *ambiguous* divide between environment and system.
 - No change in the *Hilbert* spaces.
 - No change in *unitary* evolution.
 - Just a global *consistency* constraint.
- Objective:
 - To make the theory *falsifiable*.
 - To devise an experimental test demonstrating that an isolated cyclically entangled quantum system can self-collapse.

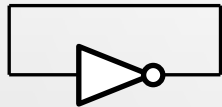
Self-Reference is Unavoidable

- $R = \{ X \mid X \notin X \}$

- $n = 1 - n$

- $n = 2 \div n$

- $z = -1/z$



- $z = +1/z$

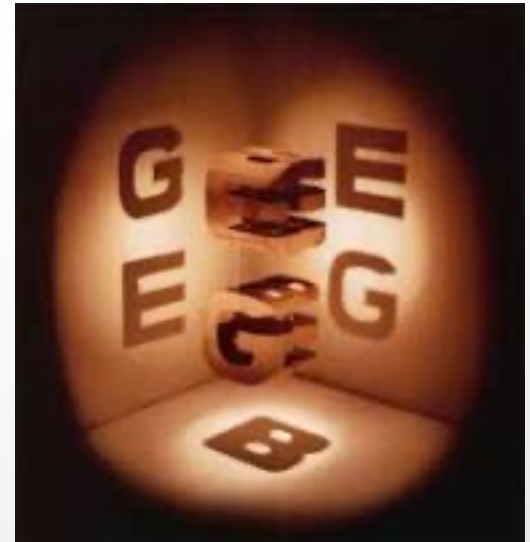


- $A : \sim A$

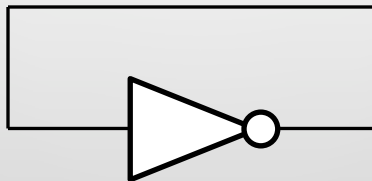
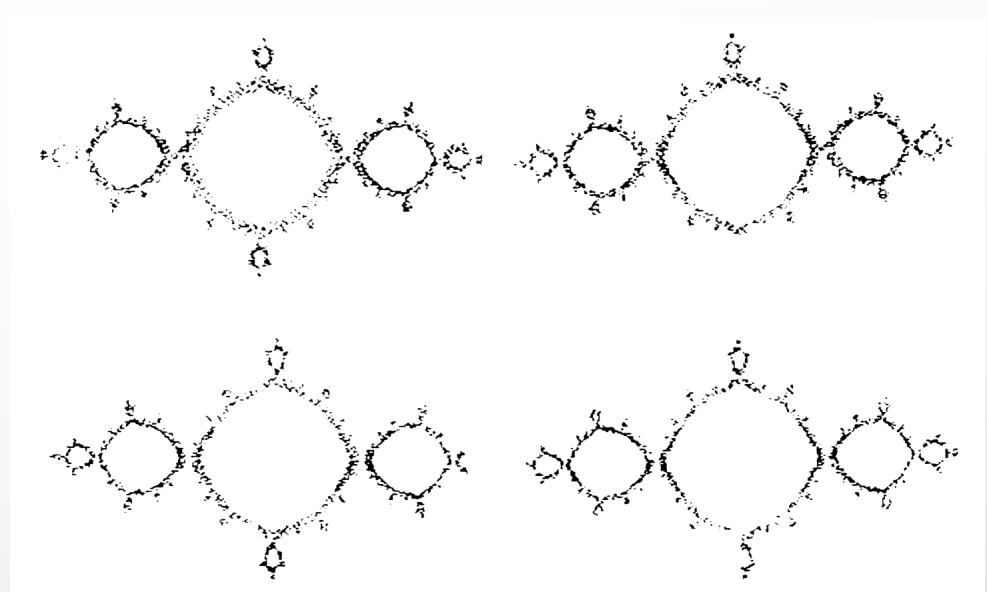
- $A : \approx A$



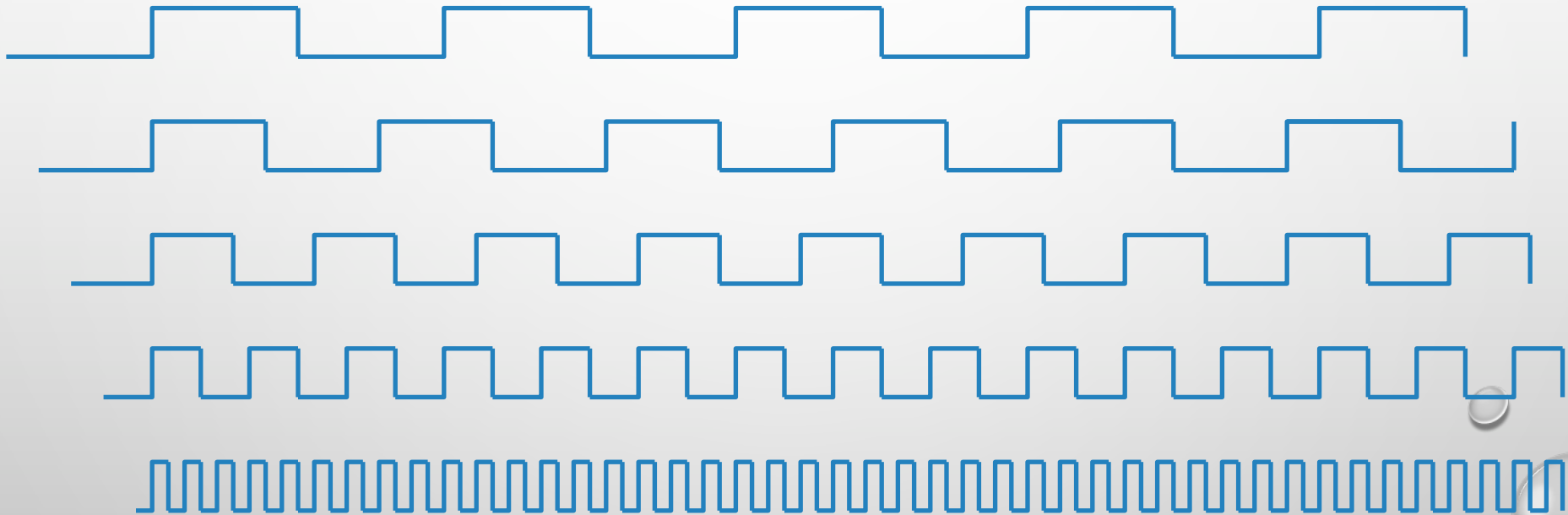
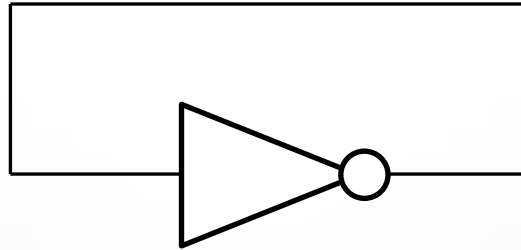
```
function P(program):  
  if Halts(program,  
    program):  
    loop forever  
  else:  
    halt  
P(P)
```



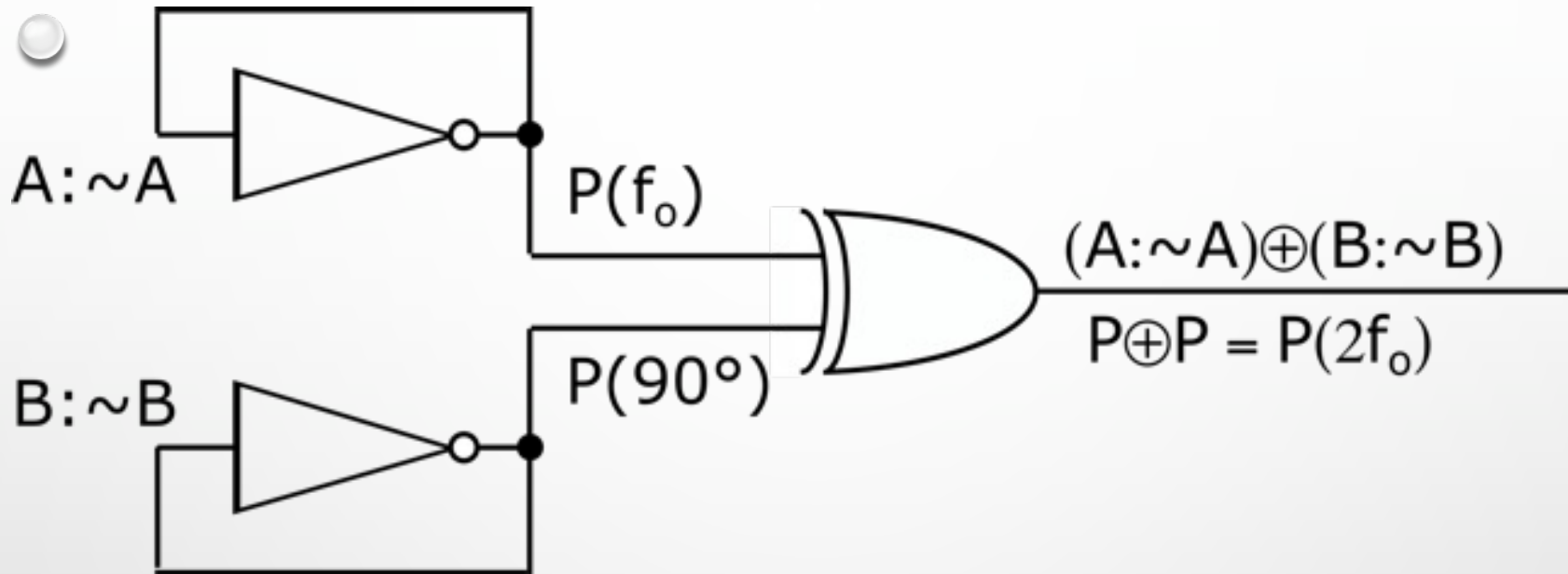
Oscillation (in spacetime?)



Frequency?



Quantization



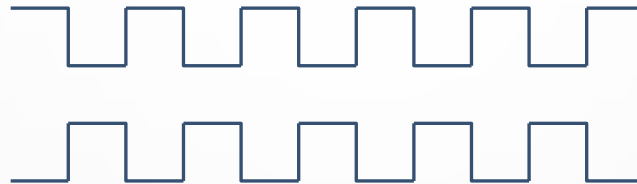
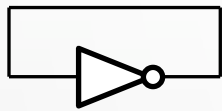
$P_A(0^\circ)$

$P_B(90^\circ)$

$P_A(0^\circ) \oplus P_B(90^\circ) = P(2f_0)$

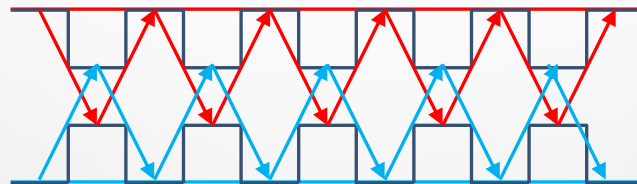


Conjugate Logical Bases



$$i = T \updownarrow F$$

$$j = F \updownarrow T$$

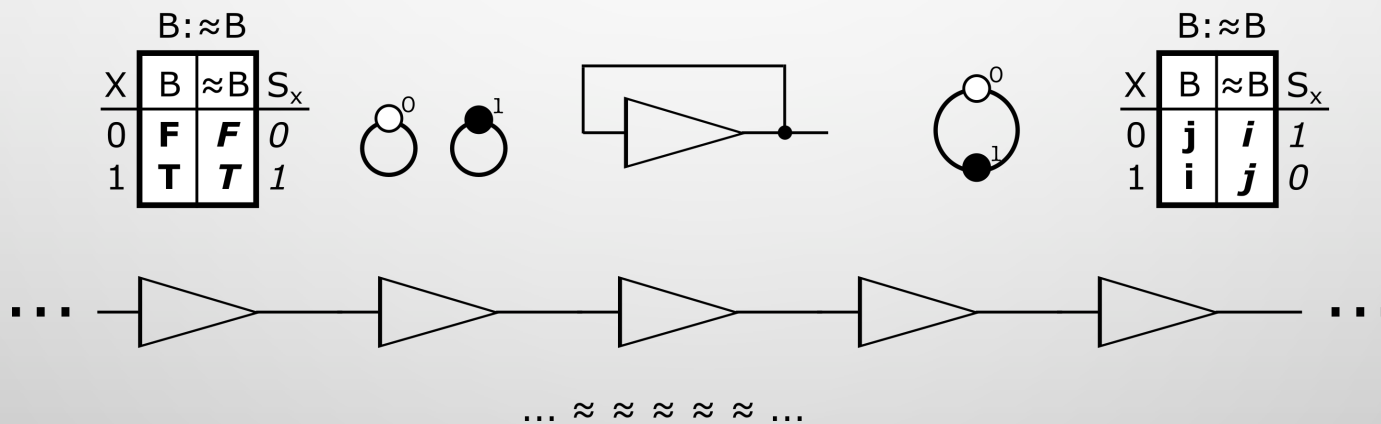
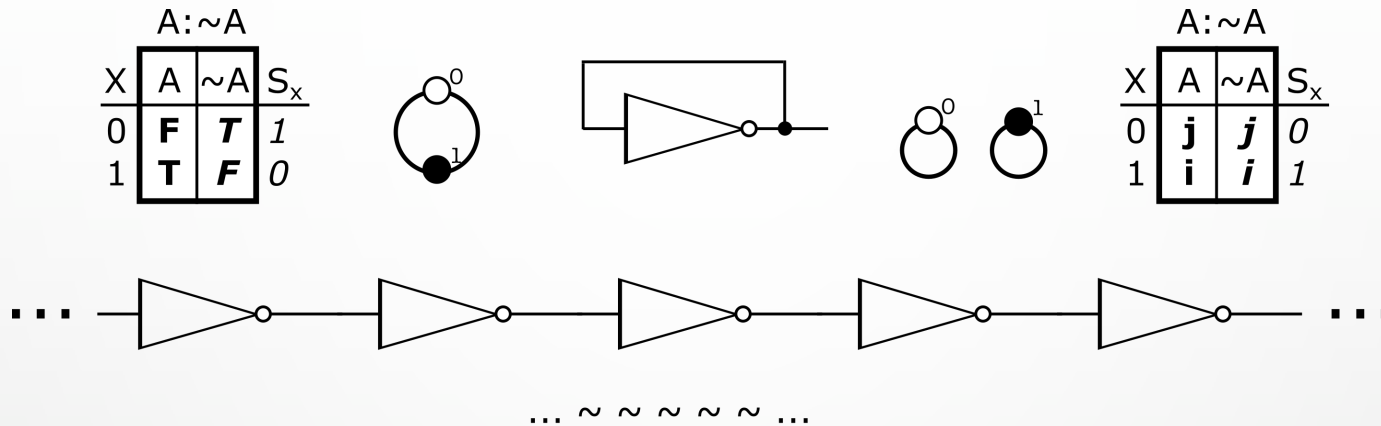


$$T = i \updownarrow j$$

$$F = j \updownarrow i$$

$$\Psi = \frac{1}{\sqrt{2}}\{|T\rangle + |F\rangle\} \quad \Phi = \frac{1}{\sqrt{2}}\{|i\rangle + |j\rangle\}$$

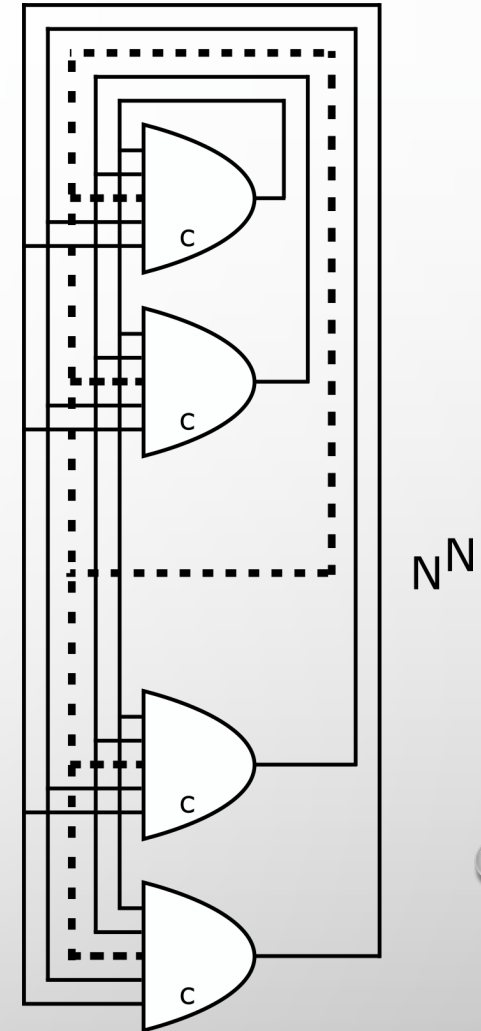
Six Representations



Scaling Law

- There are n gate complexes:
 - Each with n inputs.
 - Every output is an input.
 - Every complex is clocked.
 - Each implements a truth table.
 - Each has 2^n rows.
 - There are 2^{2^n} possible output columns.
 - Thus, $(2^{2^n})^n$ sequential circuits.
- Thus, N^N sequential circuits.
 - $N = 2^n$.
 - Only $N!$ combinatorial circuits.
- Avoiding self-reference has denied logicians access to most logical forms, $N!$ versus N^N .

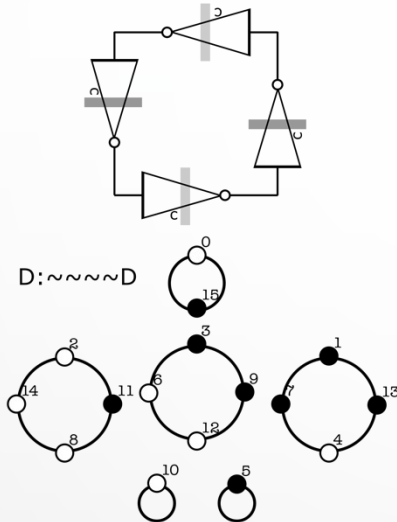
$$N = 2^n$$



Relaxation

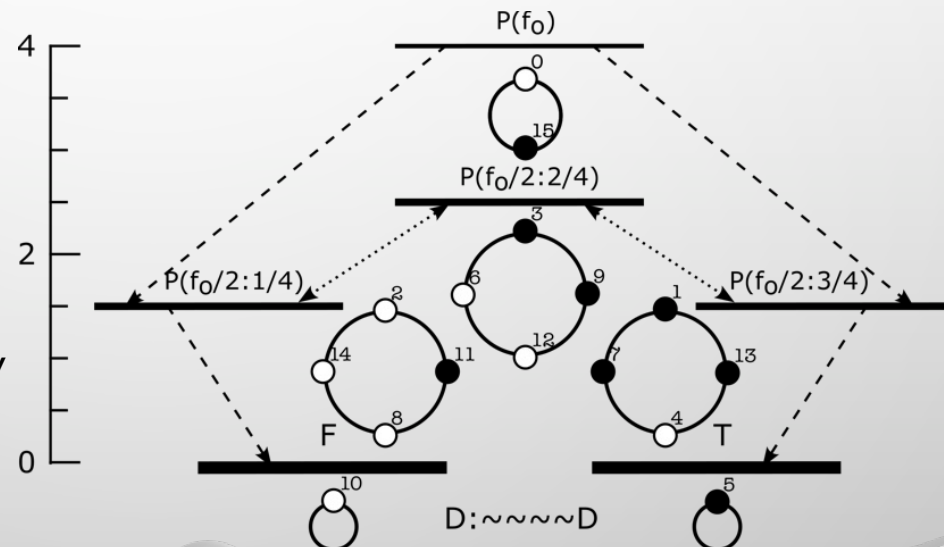
$(A:\sim B), (B:\sim C), (C:\sim D), (D:\sim A)$

X	A	B	C	D	A	B	C	D	S _x
0	F	F	F	F	T	T	T	T	15
1	F	F	F	T	T	T	F	T	13
2	F	F	T	F	T	F	T	T	11
3	F	F	T	T	T	F	F	T	9
4	F	T	F	F	F	T	T	T	7
5	F	T	F	T	F	T	F	T	5
6	F	T	T	F	F	F	T	T	3
7	F	T	T	T	F	F	F	T	1
8	T	F	F	F	T	T	T	F	14
9	T	F	F	T	T	T	F	F	12
10	T	F	T	F	T	F	T	F	10
11	T	F	T	T	T	F	F	F	8
12	T	T	F	F	F	T	T	F	6
13	T	T	F	T	F	T	F	F	4
14	T	T	T	F	F	F	T	F	2
15	T	T	T	T	F	F	F	F	0



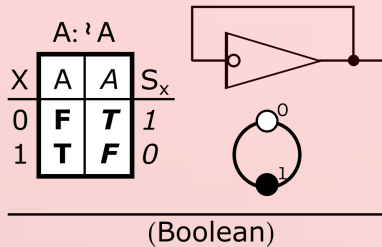
- Four NOT gates in a loop.
- Six attractors.
- Loop can contain 4, 2, or 0 contradictions.
 - 1 at the paradox frequency
 - 3 at half that
 - Different duty cycles
 - 2 fixed points

- Overclocking produces noise.
- Evaluation no longer deterministic.
- Contradictions can cancel.
 - *Irreversible state transitions.*
- Contradictions can drift.
 - *Reversible state transitions.*
- Looks a lot like excited states.

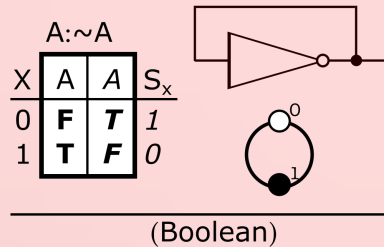


Tri Conjugate Logic (TCL)

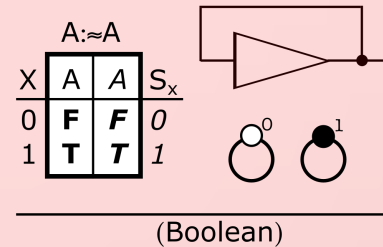
(pNOT)
1



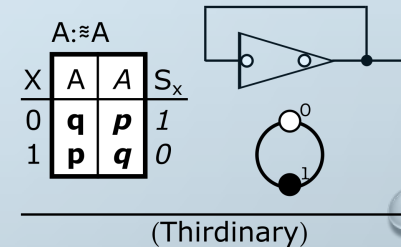
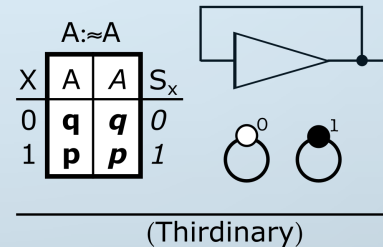
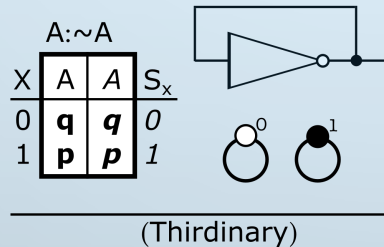
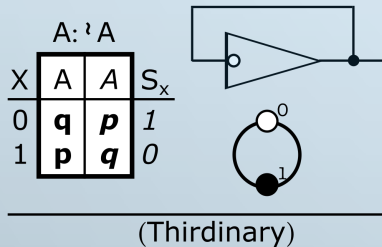
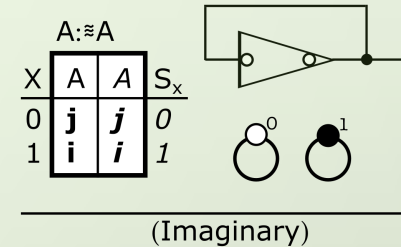
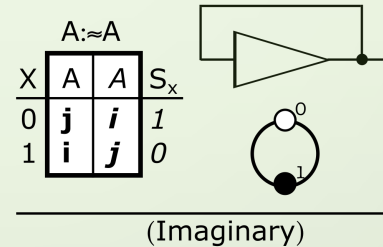
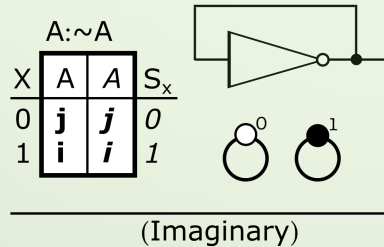
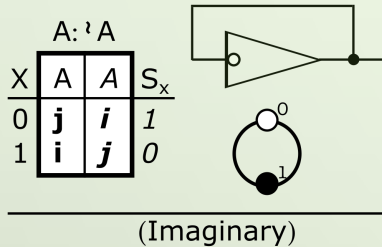
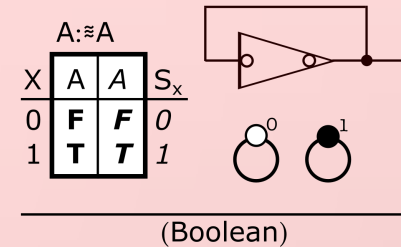
(NOT)
2



(BUF)
3



(NUF)
4

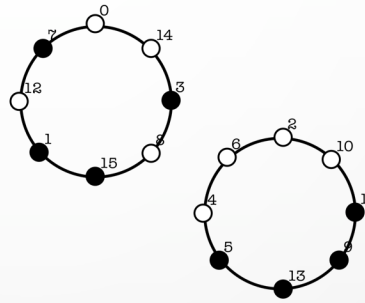
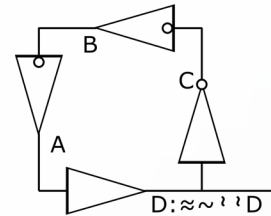


Logical Consistency

- 4 locations
- 4 operators
- 2 paradoxes
- 1 indeterminacy

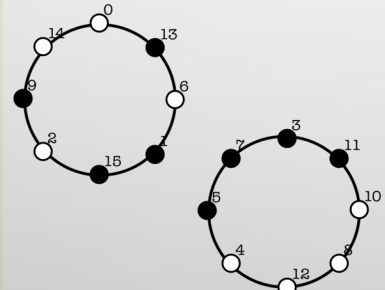
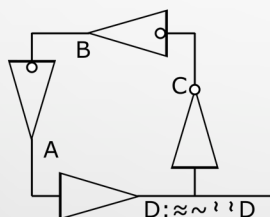
(A: 'B), (B: 'C), (C: ~D), (D: ~A)

X	A	B	C	D	A	B	C	D	S _x
0	F	F	F	F	T	T	T	F	14
1	F	F	F	T	T	T	F	F	12
2	F	F	T	F	T	F	T	F	10
3	F	F	T	T	T	F	F	F	8
4	F	T	F	F	F	T	T	F	6
5	F	T	F	T	F	T	F	F	4
6	F	T	T	F	F	F	T	F	2
7	F	T	T	T	F	F	F	F	0
8	T	F	F	F	T	T	T	T	15
9	T	F	F	T	T	T	F	T	13
10	T	F	T	F	T	F	T	T	11
11	T	F	T	T	T	F	F	T	9
12	T	T	F	F	F	T	T	T	7
13	T	T	F	T	F	T	F	T	5
14	T	T	T	F	F	F	T	T	3
15	T	T	T	T	F	F	F	T	1



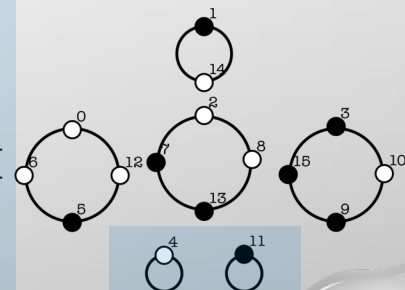
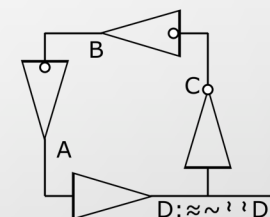
(A: 'B), (B: 'C), (C: ~D), (D: ~A)

X	A	B	C	D	A	B	C	D	S _x
0	j	j	j	j	i	i	j	i	13
1	j	j	j	i	i	i	i	i	15
2	j	j	i	j	i	j	j	i	9
3	j	j	i	i	i	j	j	i	11
4	j	i	j	j	j	i	j	i	5
5	j	i	j	i	j	i	i	i	7
6	j	i	i	j	j	j	j	i	1
7	j	i	i	i	j	j	j	i	3
8	i	j	j	j	i	i	j	j	12
9	i	j	j	i	i	j	j	j	14
10	i	j	i	j	i	j	j	j	8
11	i	j	i	i	i	j	j	j	10
12	i	i	j	j	j	i	j	j	4
13	i	i	j	i	j	i	j	j	6
14	i	i	i	j	j	j	j	j	0
15	i	i	i	i	j	j	j	j	2



(A: 'B), (B: 'C), (C: ~D), (D: ~A)

X	A	B	C	D	A	B	C	D	S _x
0	q	q	q	q	p	p	q	q	12
1	q	q	q	p	p	p	p	q	14
2	q	q	p	q	p	q	q	q	8
3	q	q	p	p	p	q	p	q	10
4	q	p	q	q	q	p	q	q	4
5	q	p	q	p	q	p	p	q	6
6	q	p	p	q	q	q	q	q	0
7	q	p	p	p	q	q	p	q	2
8	p	q	q	q	p	p	p	p	13
9	p	q	q	p	p	p	p	p	15
10	p	q	p	q	p	q	p	p	9
11	p	q	p	p	p	q	p	p	11
12	p	p	q	q	q	p	q	p	5
13	p	p	q	p	q	p	p	p	7
14	p	p	p	q	q	q	q	p	1
15	p	p	p	p	q	q	p	p	3



An Unexpected Invariant

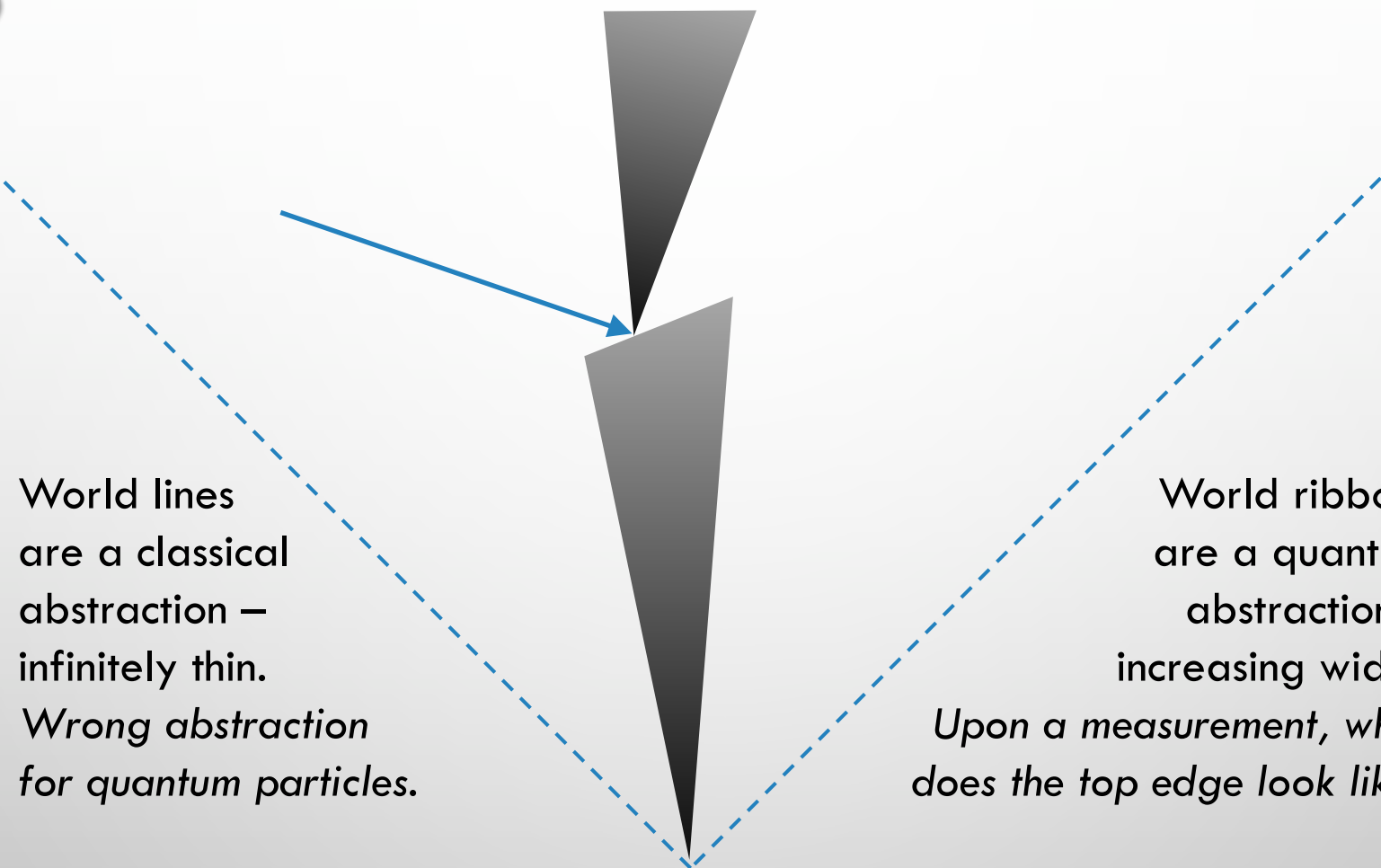
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
State	Gates				Boolean				Imaginary				Thirddinary				B	I	T	Para	0	1	2	3
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0
1	1	1	1	2	1	1	1	1	1	1	1	0	1	1	1	0	0	1	1	2	0	0	1	0
2	1	1	1	3	1	1	1	0	1	1	1	1	1	1	1	0	1	0	1	2	0	0	1	0
3	1	1	1	4	1	1	1	0	1	1	1	0	1	1	1	1	1	1	0	2	0	0	1	0
4	1	1	2	1	1	1	1	1	1	1	0	1	1	1	0	1	0	1	1	2	0	0	1	0
5	1	1	2	2	1	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	1	0	0	0
6	1	1	2	3	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	2	0	0	1	0
7	1	1	2	4	1	1	1	0	1	1	0	0	1	1	0	1	1	0	1	2	0	0	1	0
8	1	1	3	1	1	1	0	1	1	1	1	1	1	1	0	1	1	0	1	2	0	0	1	0
9	1	1	3	2	1	1	0	1	1	1	1	0	1	1	0	0	1	1	0	2	0	0	1	0
10	1	1	3	3	1	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0
11	1	1	3	4	1	1	0	0	1	1	1	0	1	1	0	1	0	1	1	2	0	0	1	0
12	1	1	4	1	1	1	0	1	1	1	0	1	1	1	1	1	1	1	0	2	0	0	1	0
246	4	4	2	3	0	0	1	0	0	0	0	1	1	1	0	0	1	1	0	2	0	0	1	0
247	4	4	2	4	0	0	1	0	0	0	0	0	1	1	0	1	1	0	1	2	0	0	1	0
248	4	4	3	1	0	0	0	1	0	0	1	1	1	1	0	1	1	0	1	2	0	0	1	0
249	4	4	3	2	0	0	0	1	0	0	1	0	1	1	0	0	1	1	0	2	0	0	1	0
250	4	4	3	3	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	1	0	0	0
251	4	4	3	4	0	0	0	0	0	0	1	0	1	1	0	1	0	1	1	2	0	0	1	0
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253	4	4	4	2	0	0	0	1	0	0	0	0	1	1	1	0	1	0	1	2	0	0	1	0
254	4	4	4	3	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	2	0	0	1	0
255	4	4	4	4	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0
																					64	0	192	0
																					25%	0%	75%	0%

The Bell States in Three Bases

State	L	O	R	Statistics
1	$\psi_- = \frac{1}{\sqrt{2}} \{ H\rangle V\rangle - V\rangle H\rangle\}$	$\psi_- = \frac{1}{\sqrt{2}} \{ D\rangle S\rangle - S\rangle D\rangle\}$	$\psi_- = \frac{1}{\sqrt{2}} \{ R\rangle L\rangle - L\rangle R\rangle\}$	Fermionic
2	$\psi_+ = \frac{1}{\sqrt{2}} \{ H\rangle V\rangle + V\rangle H\rangle\}$	$\phi_+ = \frac{1}{\sqrt{2}} \{ D\rangle D\rangle + S\rangle S\rangle\}$	$\phi_- = \frac{1}{\sqrt{2}} \{ L\rangle L\rangle - R\rangle R\rangle\}$	Bosonic
3	$\phi_- = \frac{1}{\sqrt{2}} \{ H\rangle H\rangle - V\rangle V\rangle\}$	$\psi_+ = \frac{1}{\sqrt{2}} \{ D\rangle S\rangle + S\rangle D\rangle\}$	$\phi_+ = \frac{1}{\sqrt{2}} \{ L\rangle L\rangle + R\rangle R\rangle\}$	
4	$\phi_+ = \frac{1}{\sqrt{2}} \{ H\rangle H\rangle + V\rangle V\rangle\}$	$\phi_- = \frac{1}{\sqrt{2}} \{ D\rangle D\rangle - S\rangle S\rangle\}$	$\psi_+ = \frac{1}{\sqrt{2}} \{ L\rangle R\rangle + R\rangle L\rangle\}$	

- The Bell states have the same structure as the operators of TCL.

World Ribbons



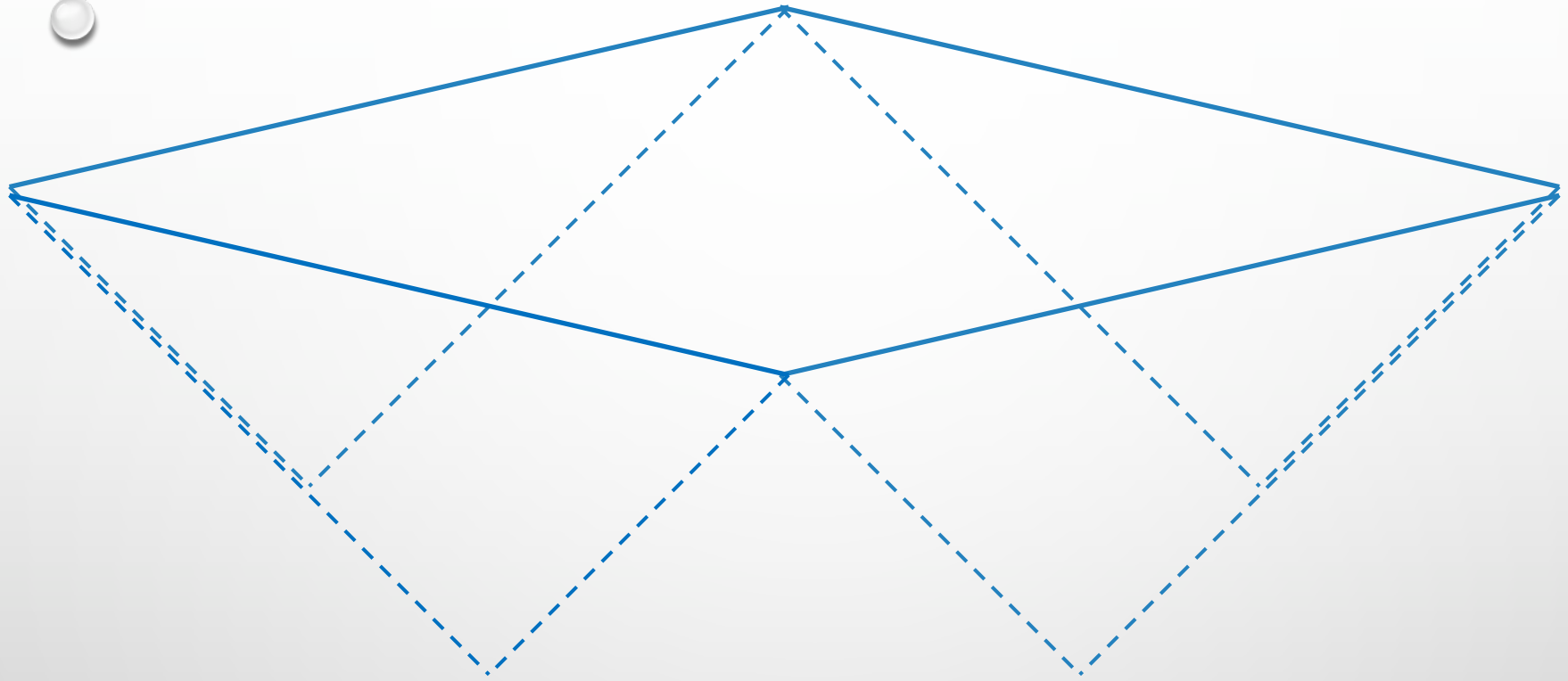
World lines
are a classical
abstraction –
infinitely thin.
*Wrong abstraction
for quantum particles.*

World ribbons
are a quantum
abstraction –
increasing width.
*Upon a measurement, what
does the top edge look like?*

A Lorentz Invariant Top Edge

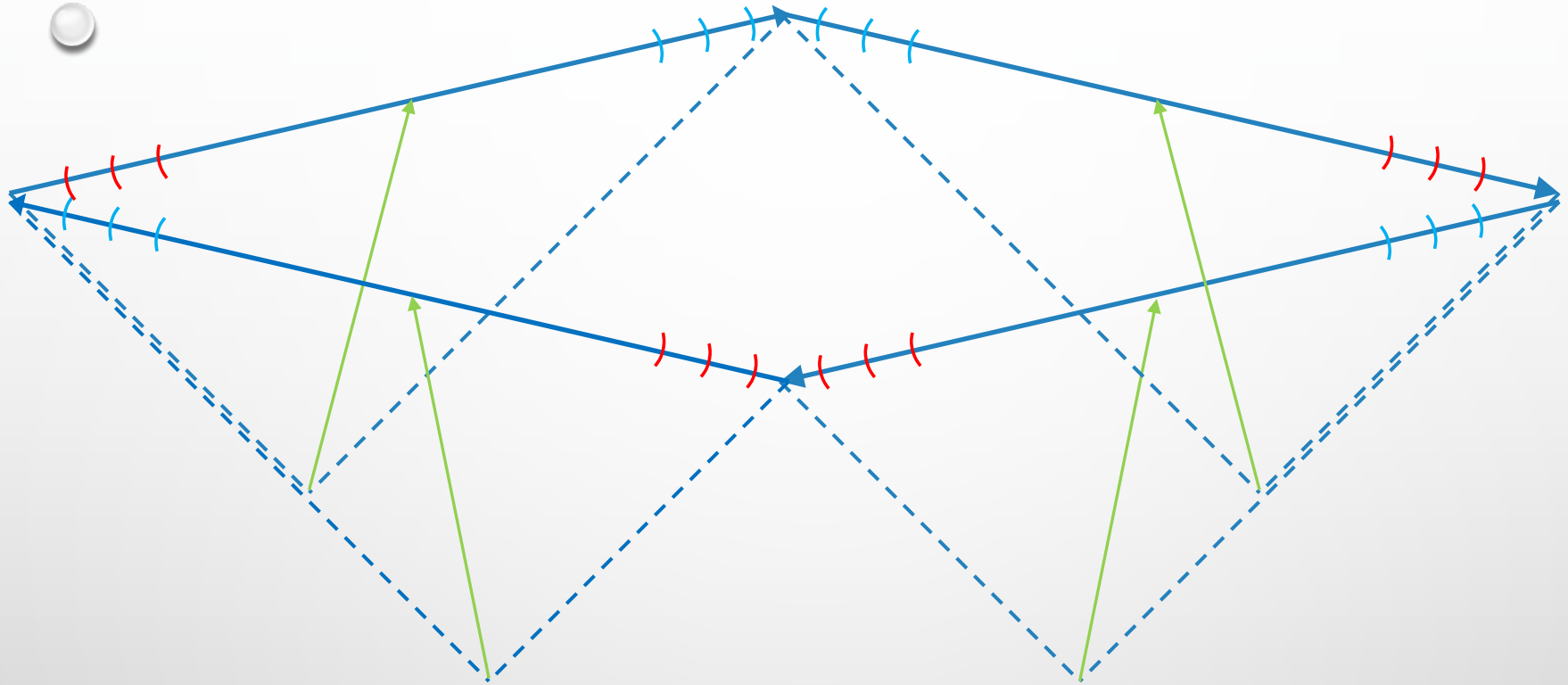
- It's called a *symmetric spacetime interval*.
- There are invariants associated with symmetric intervals – there is a firm mathematical foundation underpinning them.
- Their development is an entire presentation.
 - So, for the purposes of this talk, we simply assume them.
- Outside of self-reference, they are not testable.
- Assume that EPR pairs are coupled along symmetric intervals.
- They Lorentz transform like any other spacetime interval.
- For equal frequency pairs, the SSI is horizontal in the lab frame.
- Placing an EPR source in motion produces a Poincare rotation.
- For unequal pairs (3rd harmonic), they are steeply rotated ($\sim 0.3c$).
- Apologies for the implied “Trust Me.”

Natural Nonlinearity



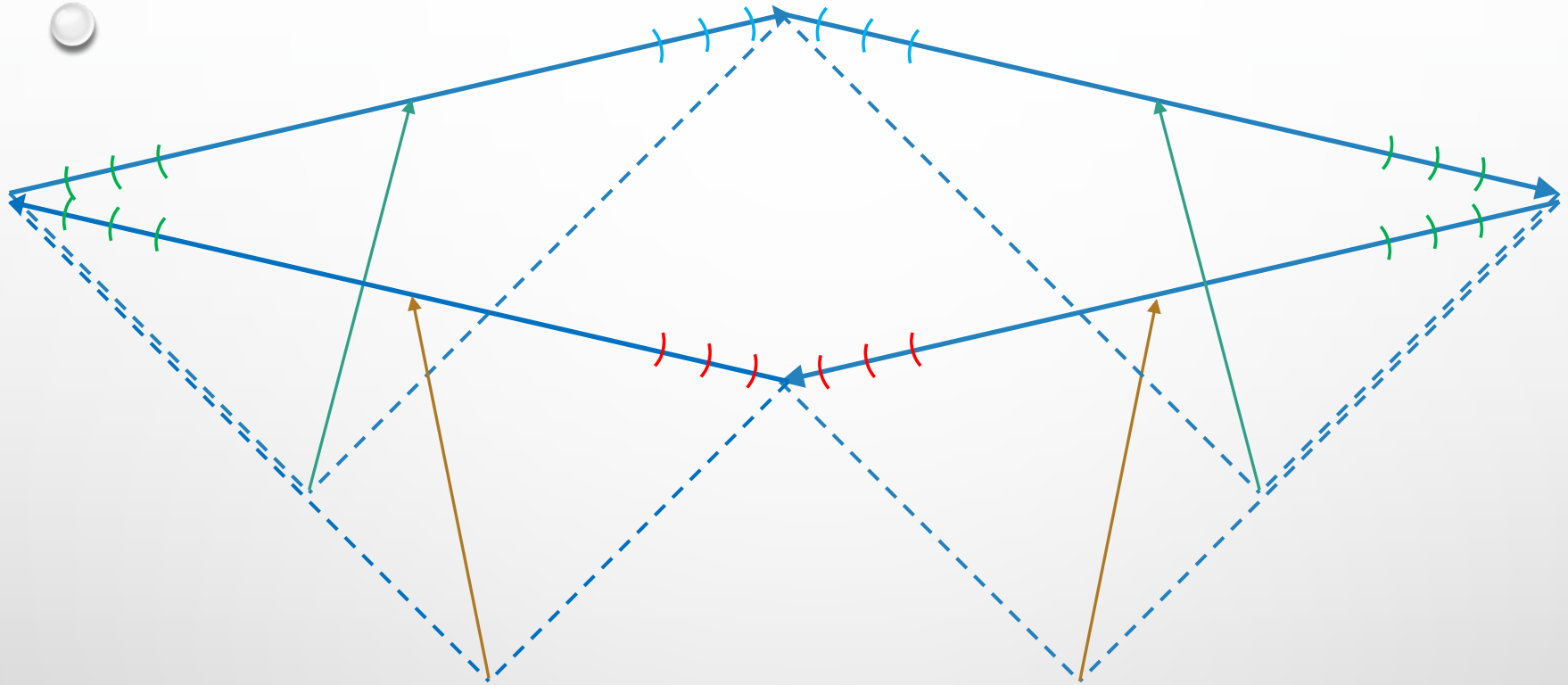
A closed loop in spacetime, consisting of four *spacelike* intervals.
Agnostic to spacelike *causality* or just spacelike *correlations*.
Proper Lorentz transformation plus Minkowski invariants.

Unitary Evolution Remains Timelike



4 EPR pairs advance into the future under standard unitary evolution.
Carefully timed to create a global consistency situation.

Bosons in the Same State



To complete the loop the cross-pair photons must be indistinguishable.
Requires they have the same frequency.
Easy to achieve by tuning the pump frequency.

An Unexpected Invariant

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
State	Pair States				Lab				O				R				L	O	R	Para	0	1	2	3
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0
1	1	1	1	2	1	1	1	1	1	1	1	0	1	1	1	0	0	1	1	2	0	0	1	0
2	1	1	1	3	1	1	1	0	1	1	1	1	1	1	1	0	1	0	1	2	0	0	1	0
3	1	1	1	4	1	1	1	0	1	1	1	0	1	1	1	1	1	1	0	2	0	0	1	0
4	1	1	2	1	1	1	1	1	1	1	0	1	1	1	0	1	0	1	1	2	0	0	1	0
5	1	1	2	2	1	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	1	0	0	0
6	1	1	2	3	1	1	1	0	1	1	0	1	1	1	0	0	1	1	0	2	0	0	1	0
7	1	1	2	4	1	1	1	0	1	1	0	0	1	1	0	1	1	0	1	2	0	0	1	0
8	1	1	3	1	1	1	0	1	1	1	1	1	1	1	0	1	1	0	1	2	0	0	1	0
9	1	1	3	2	1	1	0	1	1	1	1	0	1	1	0	0	1	1	0	2	0	0	1	0
10	1	1	3	3	1	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0
11	1	1	3	4	1	1	0	0	1	1	1	0	1	1	0	1	0	1	1	2	0	0	1	0
12	1	1	4	1	1	1	0	1	1	1	0	1	1	1	1	1	1	1	0	2	0	0	1	0
246	4	4	2	3	0	0	1	0	0	0	0	1	1	1	0	0	1	1	0	2	0	0	1	0
247	4	4	2	4	0	0	1	0	0	0	0	0	1	1	0	1	1	0	1	2	0	0	1	0
248	4	4	3	1	0	0	0	1	0	0	1	1	1	1	0	1	1	0	1	2	0	0	1	0
249	4	4	3	2	0	0	0	1	0	0	1	0	1	1	0	0	1	1	0	2	0	0	1	0
250	4	4	3	3	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	1	0	0	0
251	4	4	3	4	0	0	0	0	0	0	1	0	1	1	0	1	0	1	1	2	0	0	1	0
252	4	4	4	1	0	0	0	1	0	0	0	1	1	1	1	1	1	1	0	2	0	0	1	0
253	4	4	4	2	0	0	0	1	0	0	0	0	1	1	1	0	1	0	1	2	0	0	1	0
254	4	4	4	3	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	2	0	0	1	0
255	4	4	4	4	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0
																					64	0	192	0
																					25%	0%	75%	0%

Global Self-Consistency

- Consider circuit #6: pNOT-pNOT-NOT-BUF-...
- The truth tables in the Boolean basis:

B	A:↗B
F	T
T	F

C	B:↗C
F	T
T	F

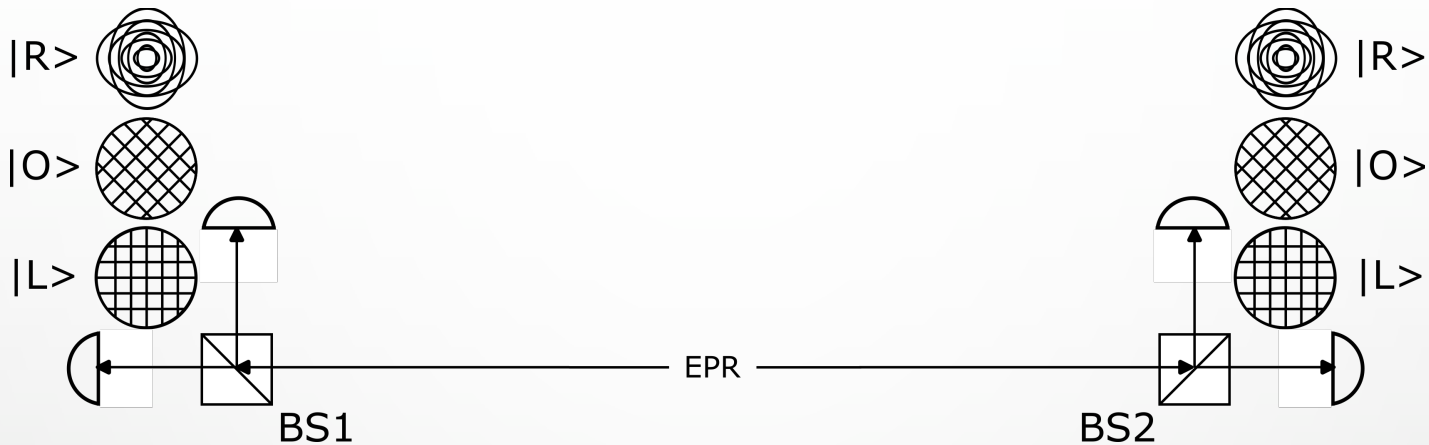
D	C:↗D
F	T
T	F

A	D:↗A
F	F
T	T

- The logical expression is $(A:↗B), (B:↗C), (C:↗D), (D:↗A)...$
- Assume D is T, then C is F, B is T, A is F which implies D is F – **contradiction**.
- Similar results in the Imaginary basis.
- Only in the *Thirday* basis are the logical values *consistent*.
- Same for permutation #6, the Bell states in the Lab basis:

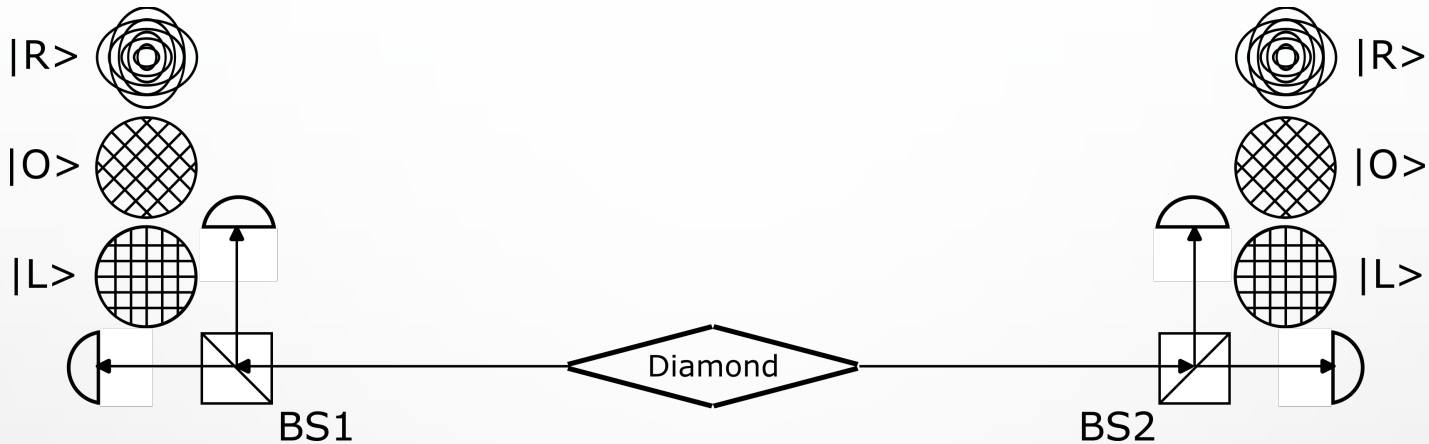
$$\psi_{-12} = \frac{1}{\sqrt{2}} \{ |H\rangle|V\rangle - |V\rangle|H\rangle \} \quad \psi_{-34} = \frac{1}{\sqrt{2}} \{ |H\rangle|V\rangle - |V\rangle|H\rangle \} \quad \psi_{+56} = \frac{1}{\sqrt{2}} \{ |H\rangle|V\rangle + |V\rangle|H\rangle \} \quad \phi_{-78} = \frac{1}{\sqrt{2}} \{ |H\rangle|H\rangle - |V\rangle|V\rangle \}$$
- The cross pairs (diamond vertices) are indistinguishable, must have the same polarization. Thus...
- Assume photons 8&1 are H, then 2&3 are V, 4&5 are H, 6&7 are V, which implies 8&1 are V – **contradiction**.
- Similar results in the O basis $|D\rangle, |S\rangle$.
- Only in the *R (Circular)* basis are the polarization states *consistent*.

EPR Entanglement



- Collapse is in the measurement basis.
- Correlations only if the bases match.
- Ambiguous Collapse Paradigm (ACP) applies.
- The *Environment* specifies all the information that determines the collapse basis.

Diamond Paradox



- Either collapse is open or occurred in the self-collapse basis.
- If collapse is open, correlations whenever the bases match.
- If self-collapse, correlations only if the bases match the self-collapse basis.
- The *quantum system* specifies all the information that determines the collapse basis.

Experimental Predictions

Collapse Measurement	Uncollapsed 25%	Lab ($ H\rangle$, $ V\rangle$)	Ortho ($ D\rangle$, $ S\rangle$)	Circular ($ L\rangle$, $ R\rangle$)
Lab ($ H\rangle$, $ V\rangle$)	Correlated	Correlated	Random	Random
Ortho ($ D\rangle$, $ S\rangle$)	Correlated	Random	Correlated	Random
Circular ($ L\rangle$, $ R\rangle$)	Correlated	Random	Random	Correlated

Falsifiable

- A speculative hypothesis deserves a good name.
- Quantum Temporal Paradox (QTP).

An Objective Measurement Mechanism

- Laws of Form led to formal self-reference which led to conjugate bases.
- Self-referential loops in logic can be *inconsistent* in some bases.
- Tri-conjugate logic (TCL) displays the same *homology* as the Bell states.
- Conjecture: quantum systems evolve until *global consistency* forces collapse.
- Spacelike connections can be arranged to be self-referential.
- Compatible with either spacelike causality or spacelike correlations.
- Diamond paradox is testable (falsifiable); makes very specific predictions.
- And it is a test of the *symmetric spacetime interval*.
 - Candidate spec for the top edge of a *world ribbon*.
- Satisfies Hawking's Chronological Protection Conjecture (CPC).
 - ***The measurement mechanism is how Nature avoids temporal paradox.***

QTP Ambitions

- QTP provides a unified candidate explanation for several open questions in quantum measurement:
 - No *ambiguous divide* between environment and quantum system.
 - Collapse basis uniquely determined by *particulars* of the quantum system.
 - Self-collapse, no *classical environment* at all – its quantum all the way down.
 - Introduces nonlinearity *naturally* through self-reference.
 - *Hilbert* spaces still apply.
 - Evolution remains *unitary*.
 - QM machinery untouched.
 - Indeterminacy provides a foundation for quantum *randomness*.
- Next steps.
 - Publish a paper on *world ribbons* and *SSI* – it is finally testable.
 - Publish a paper on *TCL*.

Q&A

Dashboard

- **Intro**

- [Argument](#)

- **Logic**

- [Self-Reference is Unavoidable](#)
 - [Oscillation](#)
 - [Frequency?](#)
 - [Quantization](#)
 - [Conjugate Logical Bases](#)
 - [Six Representations](#)
 - [Scaling Law](#)
 - [Relaxation](#)
 - [Tri Conjugate Logic \(TLC\)](#)
 - [Logical Consistency](#)
 - [An Unexpected Invariant](#)

- **Quantum Physics**

- [The Bell States in Three Basis](#)
 - [World Ribbons](#)
 - [A Lorentz Invariant Top Edge](#)
 - [Natural Nonlinearity](#)
 - [Unitary Evolution Remains Timelike](#)
 - [Bosons in the Same State](#)
 - [An Unexpected Invariant](#)
 - [Global Self-Consistency](#)
 - [EPR Entanglement](#)
 - [Diamond Paradox](#)
 - [Experimental Predictions](#)
 - [Objective Measurement Mechanism](#)
 - [QTP Ambitions](#)

- **Q&A**

- **References**

(pNOT) 1	(NOT) 2	(BUF) 3	(NUF) 4																																																
$A: \neg A$ <table border="1"> <tr><th>X</th><th>A</th><th>A</th><th>S_x</th></tr> <tr><td>0</td><td>F</td><td>T</td><td>1</td></tr> <tr><td>1</td><td>T</td><td>F</td><td>0</td></tr> </table> (Boolean)	X	A	A	S_x	0	F	T	1	1	T	F	0	$A: \sim A$ <table border="1"> <tr><th>X</th><th>A</th><th>A</th><th>S_x</th></tr> <tr><td>0</td><td>F</td><td>T</td><td>1</td></tr> <tr><td>1</td><td>T</td><td>F</td><td>0</td></tr> </table> (Boolean)	X	A	A	S_x	0	F	T	1	1	T	F	0	$A: \approx A$ <table border="1"> <tr><th>X</th><th>A</th><th>A</th><th>S_x</th></tr> <tr><td>0</td><td>F</td><td>F</td><td>0</td></tr> <tr><td>1</td><td>T</td><td>T</td><td>1</td></tr> </table> (Boolean)	X	A	A	S_x	0	F	F	0	1	T	T	1	$A: \approx A$ <table border="1"> <tr><th>X</th><th>A</th><th>A</th><th>S_x</th></tr> <tr><td>0</td><td>F</td><td>F</td><td>0</td></tr> <tr><td>1</td><td>T</td><td>T</td><td>1</td></tr> </table> (Boolean)	X	A	A	S_x	0	F	F	0	1	T	T	1
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State	L	O	R	Statistics
1	$\psi_- = \frac{1}{\sqrt{2}} \{ H\rangle V\rangle - V\rangle H\rangle \}$	$\psi_- = \frac{1}{\sqrt{2}} \{ D\rangle S\rangle - S\rangle D\rangle \}$	$\psi_- = \frac{1}{\sqrt{2}} \{ R\rangle L\rangle - L\rangle R\rangle \}$	Fermionic
2	$\psi_+ = \frac{1}{\sqrt{2}} \{ H\rangle V\rangle + V\rangle H\rangle \}$	$\phi_+ = \frac{1}{\sqrt{2}} \{ D\rangle D\rangle + S\rangle S\rangle \}$	$\phi_- = \frac{1}{\sqrt{2}} \{ L\rangle L\rangle - R\rangle R\rangle \}$	Bosonic
3	$\phi_- = \frac{1}{\sqrt{2}} \{ H\rangle H\rangle - V\rangle V\rangle \}$	$\psi_+ = \frac{1}{\sqrt{2}} \{ D\rangle S\rangle + S\rangle D\rangle \}$	$\phi_+ = \frac{1}{\sqrt{2}} \{ L\rangle L\rangle + R\rangle R\rangle \}$	
4	$\phi_+ = \frac{1}{\sqrt{2}} \{ H\rangle H\rangle + V\rangle V\rangle \}$	$\phi_- = \frac{1}{\sqrt{2}} \{ D\rangle D\rangle - S\rangle S\rangle \}$	$\psi_+ = \frac{1}{\sqrt{2}} \{ L\rangle R\rangle + R\rangle L\rangle \}$	

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Bio – Allan Goff

- Allan Goff served as a nuclear officer in the US Air Force. He has taught 50 university classes, presented 18 papers at international propulsion conferences, is the author of *Quantum Tic-Tac-Toe* and *Discovering the Rules to 3D Chess: an Unexpected Excursion into the Quantum Realm*. As an entrepreneur, he ran a placement agency for software engineers and won a two-year contract with the state of California removing 11 megawatts of demand off the electric grid. His day jobs have included software engineering and project management gigs in government, business, and academia. He is married with two children, two grandchildren...and counting.

Backup Slides

- tbd

Another Invariant

Symmetric Intervals

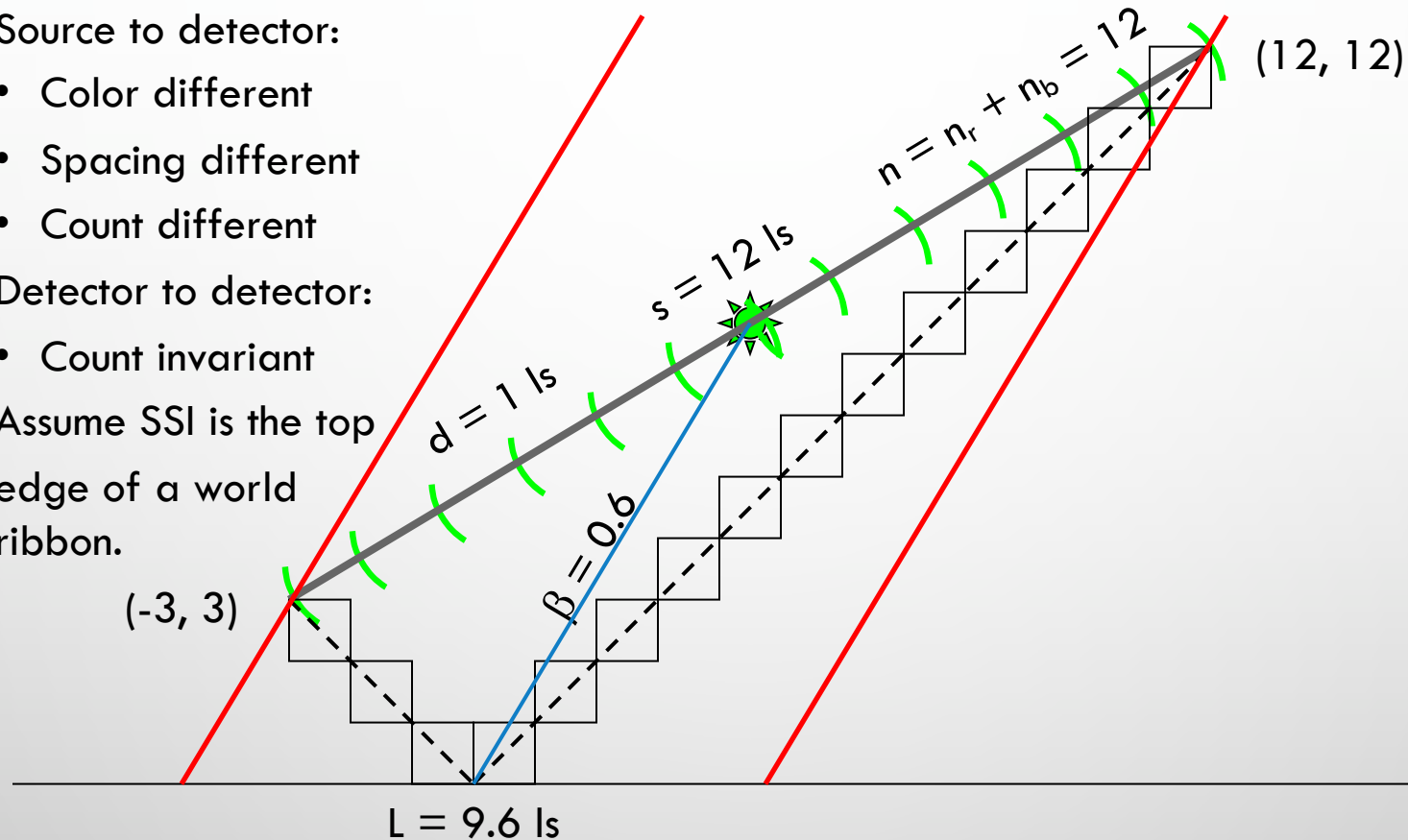
Source to detector:

- Color different
- Spacing different
- Count different

Detector to detector:

- Count invariant

Assume SSI is the top edge of a world ribbon.



Global Self-Consistency - TCL

- Consider circuit #6: pNOT-pNOT-NOT-BUF-...
- The truth tables in the Boolean basis:

B	$A \rightarrow B$	C	$B \rightarrow C$	D	$C \sim D$	A	$D \approx A$
F	T	F	T	F	T	F	F
T	F	T	F	T	F	T	T

- The logical expression is $(A \rightarrow B), (B \rightarrow C), (C \sim D), (D \approx A) \dots$
- Assume D is T, then C is F, B is T, A is F which implies D is F – **contradiction**.
- Similar results in the Imaginary basis.
- Only in the Thirdinary basis are the logical values consistent.

Global Self-Consistency – Bell

- Consider loop #6:
- The Bell states in the Lab basis:
- $\psi_{-12} = \frac{1}{\sqrt{2}} \{ |H\rangle|V\rangle - |V\rangle|H\rangle \}$
- $\psi_{-34} = \frac{1}{\sqrt{2}} \{ |H\rangle|V\rangle - |V\rangle|H\rangle \}$
- $\psi_{+56} = \frac{1}{\sqrt{2}} \{ |H\rangle|V\rangle + |V\rangle|H\rangle \}$
- $\phi_{-78} = \frac{1}{\sqrt{2}} \{ |H\rangle|H\rangle - |V\rangle|V\rangle \}$
- The cross pairs (diamond vertices) are indistinguishable, must have the same polarization. Thus...
- Assume photons 8&1 are H, then 2&3 are V, 4&5 are H, 6&7 are V, which implies 8&1 are V – **contradiction**.
- Similar results in the O basis $|D\rangle$, $|S\rangle$.
- Only in the R basis are the polarization states *consistent*.

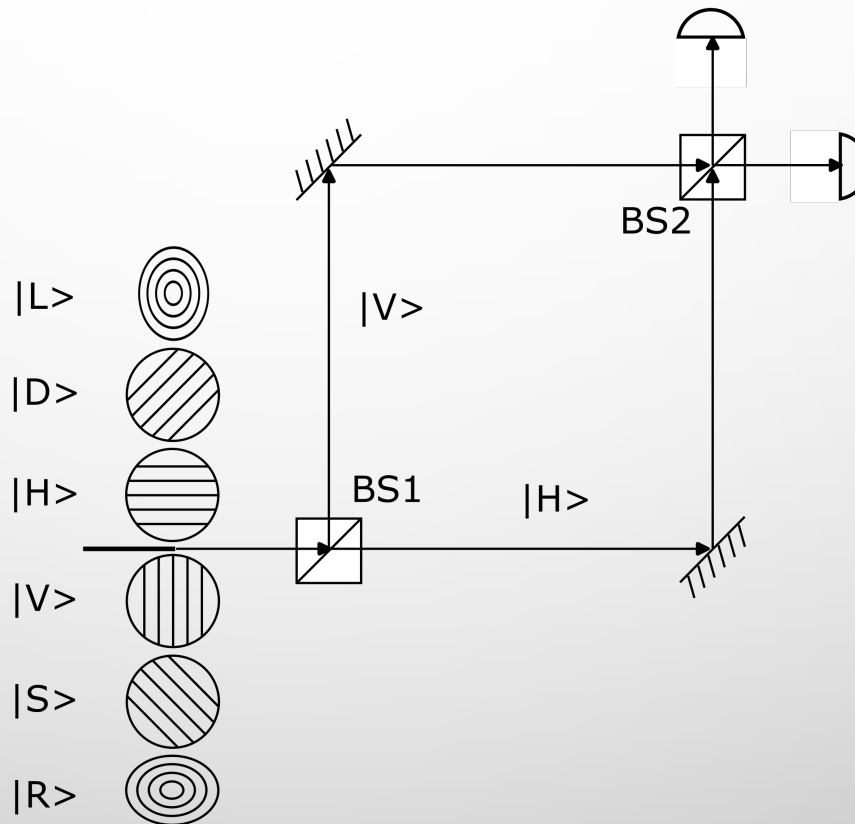
Diffraction

Mach-Zehnder interferometer in the lab basis.

One arm is $|H\rangle$, one is $|V\rangle$.

Measure in a basis.

Diffraction patterns only.

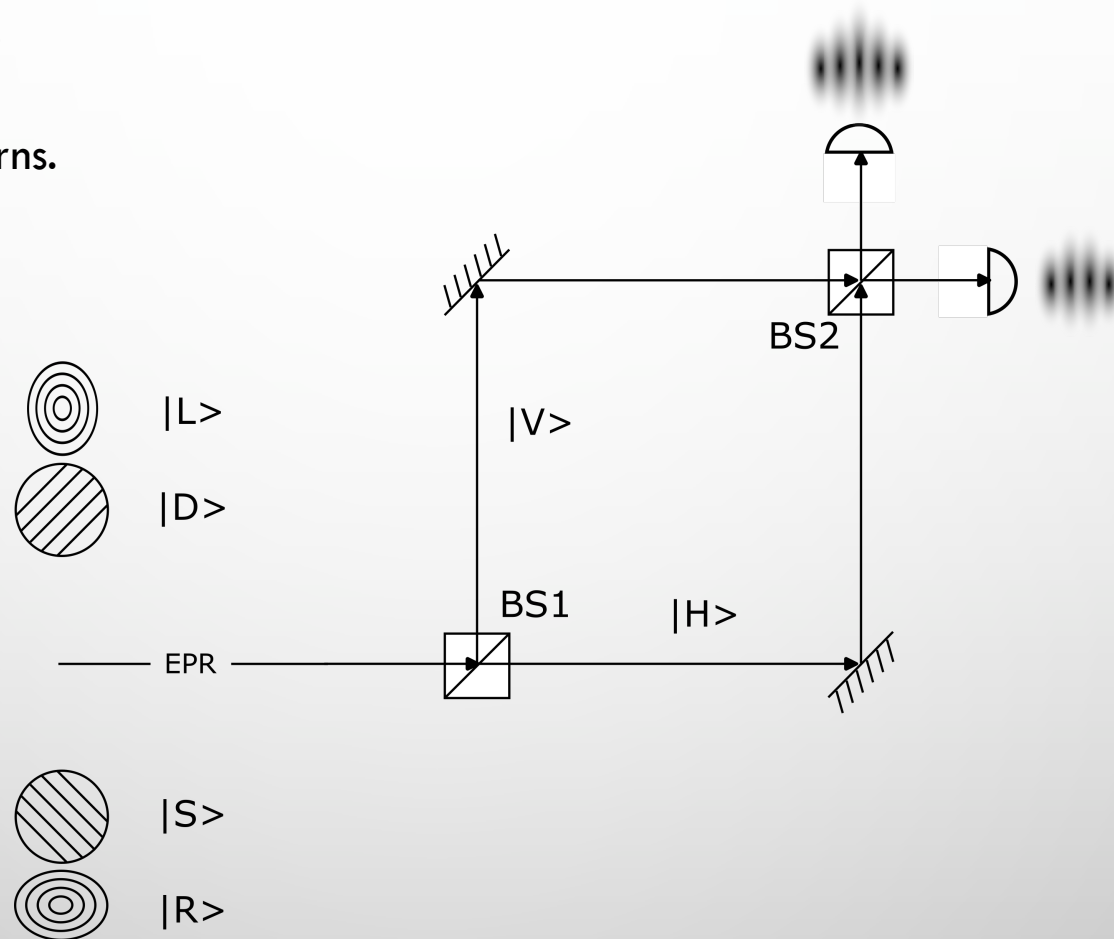


Interference

Use entanglement.

Orthogonal bases.

Post-select to see
interference patterns.

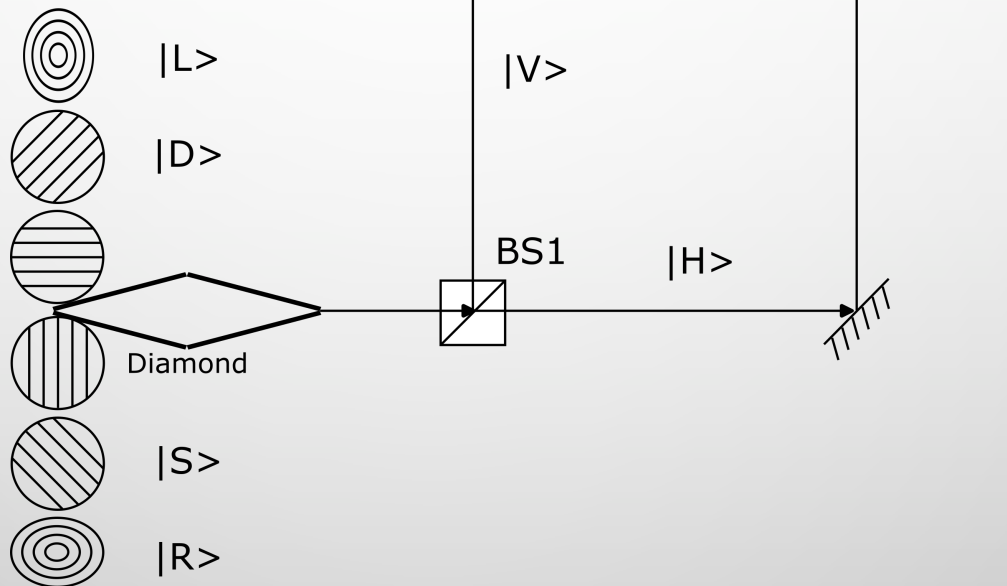


An Experimental Prediction

Test diamond loop.

Interference only for indeterminate loops in the matching basis.

The claim of self-collapse is *testable*.



Experimental Permutations

Collapse Measurement	Uncollapsed 25%	Lab ($ H\rangle, V\rangle$)	Ortho ($ D\rangle, S\rangle$)	Circular ($ L\rangle, R\rangle$)
Lab ($ H\rangle, V\rangle$)	Diffraction	Diffraction	Diffraction	Diffraction
Ortho ($ D\rangle, S\rangle$)	Interference	Diffraction	Interference	Diffraction
Circular ($ L\rangle, R\rangle$)	Interference	Diffraction	Diffraction	Interference

Falsifiable

- QTP creates tension with our classical notion of linear time.
 - The present as an infinitely thin interface between past and future.
- The diamond loop is a static structure in spacetime.
- Any photon prior to encountering the loop is indeterminate.
- After a collapsed loop, every photon's state is fixed.
- Have entangled the near future with the recent past into an expanded present.
- Within this window around *now*, past, present, and future become ambiguous.
- The present has a *thickness* – call it a *chronoblock*.

An Objective Measurement Mechanism?

- Laws of Form led to formal self-reference which led to conjugate bases.
- Self-referential loops in logic can be *inconsistent* in some bases.
- Tri-conjugate logic (TCL) displays the same *homology* as the Bell states.
- Conjecture: quantum systems evolve until *global consistency* forces collapse.
- Spacelike connections can be arranged to be self-referential.
- Compatible with either spacelike causality or spacelike correlations.
- Diamond paradox is testable (falsifiable); makes very specific predictions.
- However, it falls *short* of QTP ambitions.
 - The MZI provides symmetry breaking, the basis of the arms.
 - In an EPR experiment, the environment *fully* defines the collapse basis.
 - In the diamond paradox the MZI is not sufficient, the loop provides *part* of the specification.
 - A step closer to self-collapse – quantum level only.
- But it is a test of the *symmetric spacetime interval*.
 - Candidate spec for the top edge of a *world ribbon*.

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